# Weekly Retrieval-Practice Quizzes and Final Exam Performance in Introductory Statistics: A Cluster-Randomised Trial

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## Abstract

**Background.** Retrieval practice improves retention in laboratory studies, but classroom evidence is mixed. **Methods.** Twelve sections of an introductory statistics course (418 students) were randomised 1:1 to weekly low-stakes quizzes or to re-reading assignments. **Results.** Students who completed most quizzes scored 4.8 points higher on the final exam (p = .004, n = 247). **Conclusions.** Weekly quizzing significantly improves exam performance and reduces statistics anxiety, particularly for students with weaker preparation.

## 1. Introduction

Testing oneself on studied material, known as retrieval practice, strengthens long-term memory more than re-reading the same material. Most of this evidence comes from short laboratory sessions with word lists or prose passages. Whether regular low-stakes quizzes improve performance in a full-semester university course, where students choose how much to engage, is less clear. We tested weekly quizzes in a large introductory statistics course.

## 2. Methods

### 2.1 Design and participants

We conducted a cluster-randomised trial in the 2024-25 academic year. The unit of randomisation was the course section: 12 sections, each taught by a different instructor, were allocated 1:1 by a computer-generated sequence to the quiz arm (6 sections) or the control arm (6 sections). All 418 students enrolled in these sections were eligible and included.

### 2.2 Intervention

Quiz sections completed ten weekly online quizzes of eight short-answer items with feedback. Control sections received ten weekly re-reading assignments covering the same textbook pages. Neither activity counted toward the course grade.

### 2.3 Outcomes

The pre-registered primary outcome was the final exam score (0-100), marked blind to arm. Secondary outcomes were the midterm score, statistics anxiety measured with the STARS questionnaire (total score 23-115) at the end of term, and course withdrawal.

### 2.4 Statistical analysis

The primary analysis compared arms on the final exam score with a linear mixed model with a random intercept for section, including all randomised students (intention-to-treat); missing final exam scores were multiply imputed (20 imputations). The three secondary outcomes were tested at a Bonferroni-adjusted significance level of alpha = 0.0167 (0.05/3). No subgroup analyses were pre-specified. A per-protocol analysis restricted to students who completed at least 8 of the 10 assigned activities was planned as a sensitivity analysis only.

## 3. Results

### 3.1 Participants

Of the 418 enrolled students (quiz arm n = 209; control arm n = 209), 391 sat the final exam (quiz arm n = 196; control arm n = 195). Figure 1 shows the flow of sections and students.

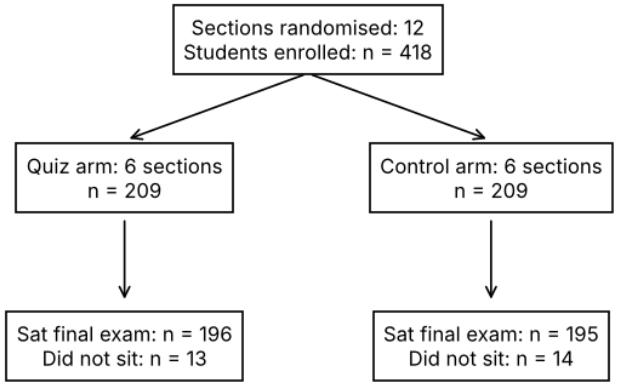



**Figure 1.** Flow of course sections and students through the trial.

## 3.2 Primary outcome

In the intention-to-treat analysis, the adjusted mean difference in final exam score between the quiz and control arms was 2.1 points (95% CI -0.4 to 4.6; p = 0.098; n = 418). Figure 2 shows the distribution of observed scores in each arm.

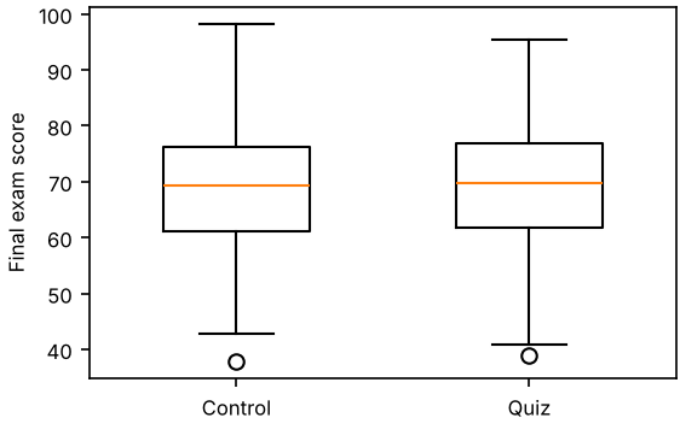



**Figure 2.** Observed final exam scores by arm among students who sat the exam.

## 3.3 Per-protocol analysis

Among students who completed at least 8 of the 10 assigned activities (n = 247), the difference in final exam score was 4.8 points (95% CI 1.6 to 8.0; p = .004).

## 3.4 Secondary outcomes

Midterm scores were 1.6 points higher in the quiz arm (p = 0.21; n = 405). End-of-term STARS anxiety was 2.3 points lower in the quiz arm (p = 0.031; n = 377). Course withdrawal did not differ between arms (odds ratio 0.71; p = 0.27; n = 418).

## 3.5 Exploratory analysis by prior mathematics grade

Table 2 reports the final exam difference separately for students with low and high prior mathematics grades. This analysis was not pre-specified.

**Table 2.** Final exam score difference (quiz minus control) by prior mathematics grade, students who sat the final exam.

| Subgroup | n | Difference in final exam score (95% CI) | p |
|---|---|---|---|
| Low prior grade (C or below) | 141 | 5.9 (1.3 to 10.5) | 0.012 |
| High prior grade (B or above) | 250 | 0.3 (-2.8 to 3.4) | 0.84 |

## 3.6 Quiz engagement

Engagement with the weekly quizzes fell over the term; Figure 3 shows the share of quizzes submitted in each two-week block.

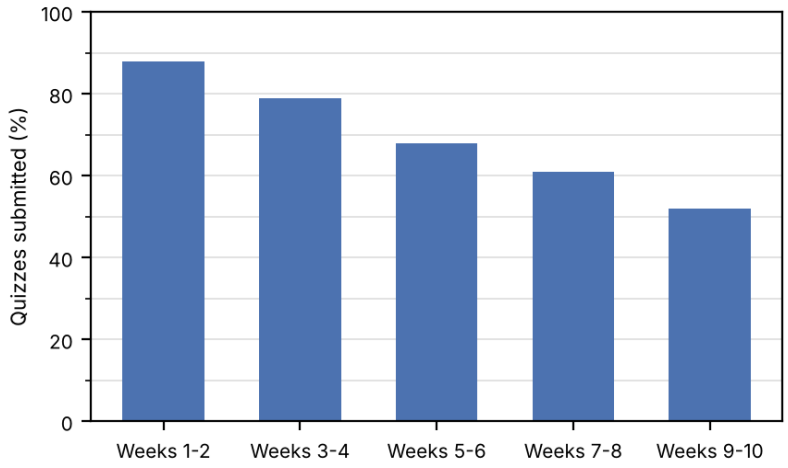



**Figure 3.** Weekly quiz completion in the quiz arm by two-week block (percentage of quizzes due in the block that were submitted). Completion declined across the term (test for linear trend, P < 0.001; n = 209).

## 4. Discussion

This trial shows that weekly retrieval-practice quizzes significantly improve final exam performance in introductory statistics, raising scores by almost five points. Quizzing also significantly reduced statistics anxiety, a common barrier in required methods courses. The benefit was concentrated among students with weaker prior preparation (Table 2), which suggests that instructors should target quizzes at this group. Because students were individually randomised, the differences between arms cannot be attributed to instructor effects.

Limitations include the single institution, the self-reported anxiety measure, and the declining quiz engagement over the term (Figure 3).

## Appendix A. Sensitivity analysis

A complete-case analysis of the final exam score without imputation gave a difference of 2.4 points (95% CI -0.2 to 5.0; p = 0.071; n = 391).

## Appendix B. Baseline balance

Prior mathematics grade differed between arms at baseline (p = 0.046; n = 409 students with transcript records). The primary model did not adjust for prior grade.